Recipe Helper

Intelligent Multimodal Recipe Assistant Based on Agentic AI and RAG

From a simple text request or food photo to a personalized, verified recipe

RECOGNIZE INGREDIENTS

RETRIEVE RECIPES

VERIFY ANSWERS



The search box assumes you already know the answer

Millions of recipes exist. Personal context is the missing ingredient.

- Fridge photo, but no dish name
- Dish photo, but no recipe
- Documents in other languages or formats
- Dietary, time, and calorie constraints

Keyword search cannot reason across these inputs. Recipe Helper can.



Formal Problem Statement

“Given a multimodal user request plus optional constraints, retrieve the most relevant recipe from a large corpus and present it with verified, safe, personalized guidance.”

MULTIMODALITY	SCALE	GROUNDING	ORCHESTRATION
Text, ingredient photo, dish photo, or recipe document	Semantic retrieval across 221,741 indexed recipes	Control hallucinations in generated content	Coordinate multiple specialized language models

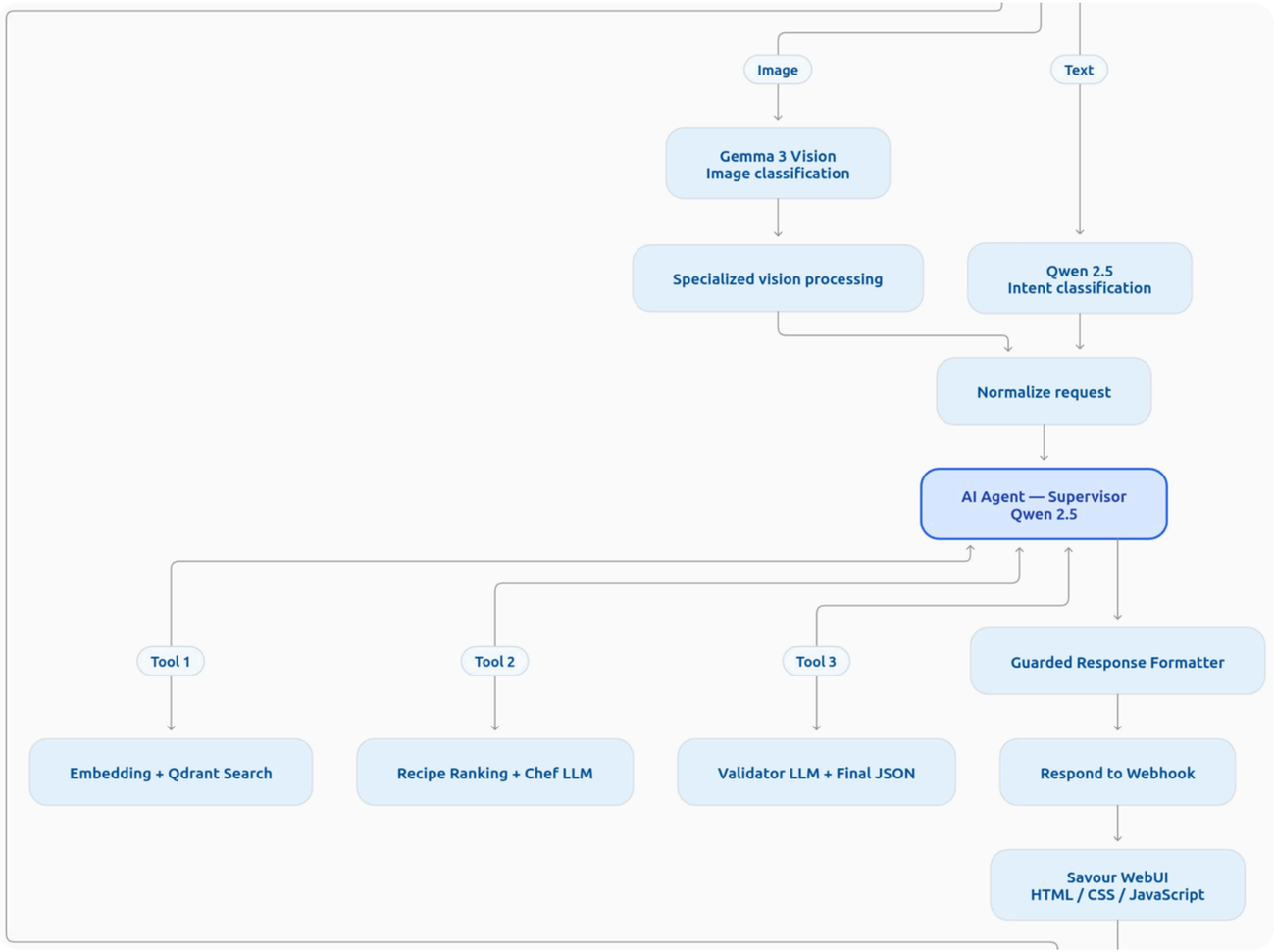
221,741 recipes — structured, filterable, inspectable

221,741

RECIPES INDEXED

`name`
`ingredients · ingredient_lines`
`steps · minutes · cuisine_type`
`calories · protein · carbs · fat`
`vegetarian · image_url`

Architecture evidence shown at right; production payloads preserve authoritative recipe records and filterable metadata.



Data preparation makes retrieval trustworthy

01 NORMALIZE

Ingredient names

Singular forms and consistent naming. Salt, water, and oil classified as pantry basics.

02 STRUCTURE

Filterable payload

Minutes, calories, and protein stored as numeric Qdrant payload filters.

03 PRECOMPUTE

Vegetarian flag

A trusted hard filter—rather than a language-model guess at query time.

Indexing at scale, across languages

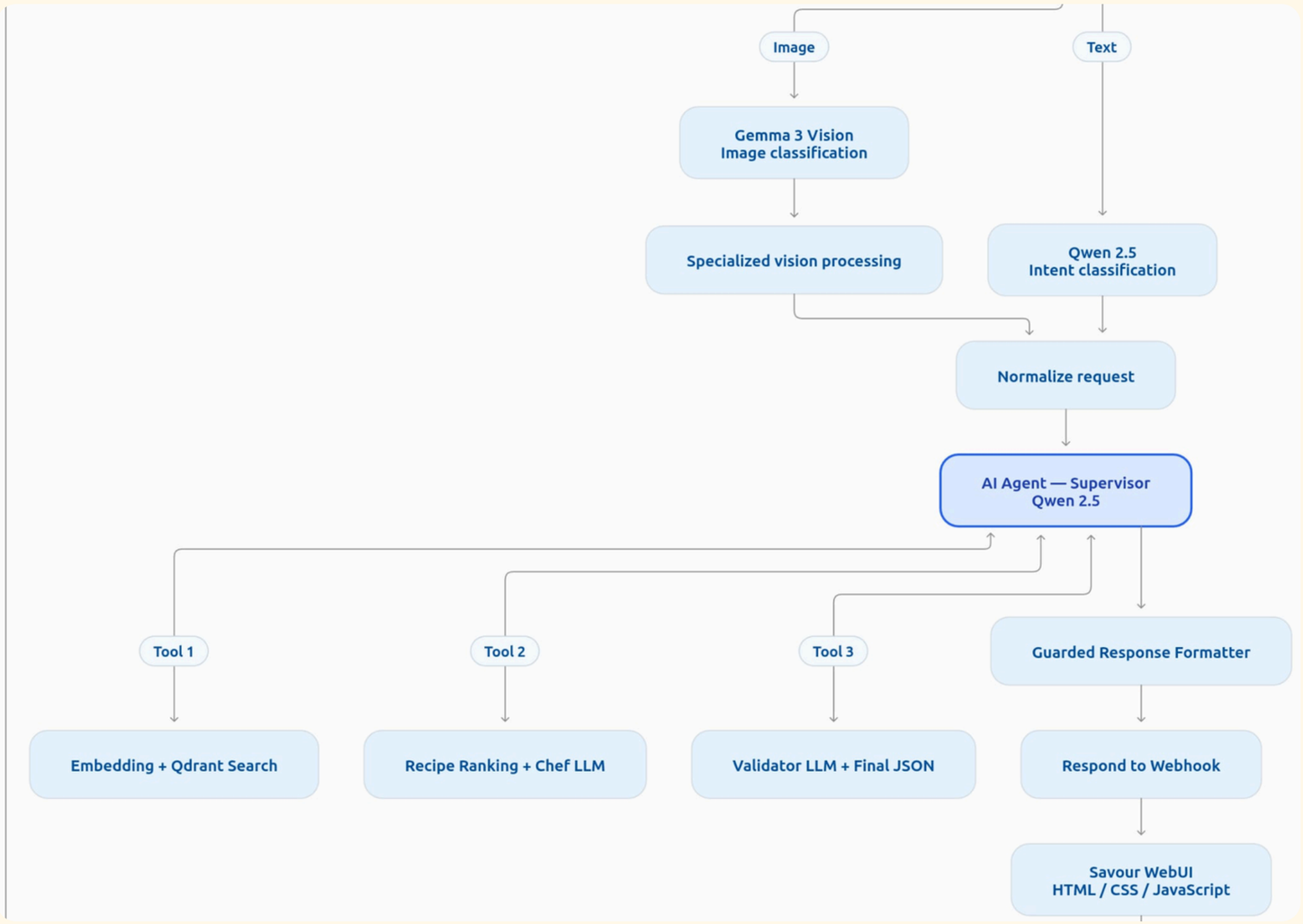


221,741

recipe vectors

Payload-indexed filters narrow the search before ranking—efficient retrieval without giving up semantic similarity.

One workflow. Three embedded tools. A supervised system.



The tools are inline n8n workflow definitions—not three separately deployed workflows.

VISION + CHEF
Gemma 3 4B

INTENT + SUPERVISOR + VALIDATOR
Qwen 2.5 3B

RETRIEVAL
BGE-M3 + Qdrant

RUNTIME
Savour WebUI · Ollama · Docker

Two branches, one normalized request

IMAGE BRANCH

Gemma 3 4B sees; it does not invent.

Classifies: ingredients_photo · dish_photo · recipe_document · not_food

Extracts visible ingredients, identifies dishes, or parses documents. Confidence thresholds route uncertain detections safely.

TEXT BRANCH

Qwen 2.5 3B turns intent into structure.

Classifies: ingredients · dish · help

Extracts requested dishes, ingredient lists, and constraints. Supports multilingual French and Arabic input.

From kitchen line to chef supervisor

FIXED PIPELINE

Search → Rank → Chef → Validator → Response

Every request follows the same sequence. Errors propagate forward; revision is awkward.

AGENTIC SUPERVISION

A supervisor dispatches the right station.

Chooses tools dynamically, controls arguments, retries or revises, respects iteration limits, and verifies output before service.

Technology stack: responsibility by layer

EXPERIENCE *Savour WebUI* — interaction, images, preferences, display

ORCHESTRATION n8n — webhooks, routing, embedded tools

INTELLIGENCE Gemma 3 4B — vision + Chef · Qwen 2.5 3B — intent + supervision + validation

RETRIEVAL BGE-M3 — embeddings · Qdrant — vector search and payloads

RUNTIME Ollama — local inference · Docker — service deployment

Local AI is a control choice—not a magic offline switch

CONTROL

Deployment environment
No per-request external API charges
Reduced inference dependence

COST

Hardware requirements
Latency considerations
Smaller capacity and context windows

Caution: claim full offline operation only when every component actually runs locally.

Retrieval is a six-course sequence

1
Normalize multimodal request

2
Embed with BGE-M3

3
Search Qdrant + filters

4
Fetch payloads by ID

5
Rank relevance + overlap

6
Chef adds grounded guidance

Ingredient mode deliberately favors recipes sharing visible ingredients—even when pure semantic similarity ranks another dish higher.

Prompt engineering by role

VISION CLASSIFIER	One category + confidence · JSON
INGREDIENT EXTRACTOR	Never invent unseen ingredients
CHEF	Only grounded intro, tips, substitutions
VALIDATOR	Reject unsupported or unsafe content
SUPERVISOR	Honor tool contract, limits, recipe IDs

`JSON schemas · low-temperature extraction · few-shot examples · negative instructions`

Shrink the hallucination surface

QDRANT — AUTHORITATIVE
Title · ingredients · steps · nutrition

CHEF — SUPPLEMENTARY
Commentary · tips · valid substitutions

The final recipe is built from database facts first. Generation is deliberately confined to helpful context.

Known limitation: ingredient checks can catch erroneous vegetarian flags, but current rules do not detect every animal-derived ingredient.

The Supervisor coordinates records—it does not author them

Sub_Search_Embed

Embedding + Qdrant search

Returns candidate IDs, previews, search status.

Sub_Rank_Chef

Payload retrieval + ranking + Chef

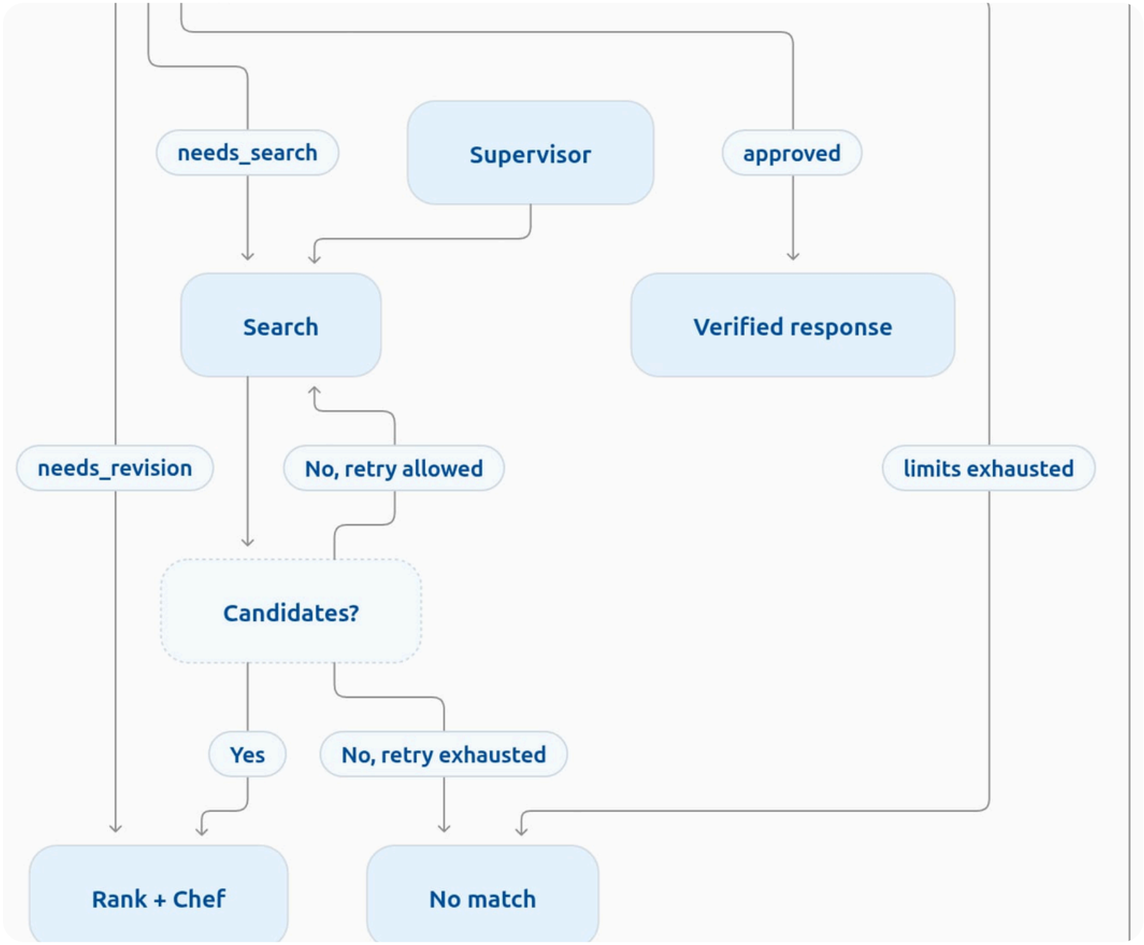
Returns selected ID, draft, matched and missing ingredients.

Sub_Validate_Final_Build

Recipe verification + Validator + final build

Returns approval, issues, safe draft or final recipe.

A bounded decision loop—not an infinite tasting session



RETRY CONDITIONS

No candidates
Weak match
Validation issue
Missing required status

Each retry targets a specific failure. Identical repeated calls are avoided; iteration limits force a safe stop.

Reliability comes from explicit contracts

1 Trusted request context

2 Authoritative retrieval by ID

3 Independent validation

4 Guarded final formatting

DEBUGGING NOTE

The Supervisor once repeated Search and Chef because the Validator lacked the selected recipe ID.

Fix: pass chosen_id explicitly to Validator. The loop then gained the state needed to approve or reject the same record.

Chef’s tasting notes: what the system taught us

1 / Multimodal + multilingual input can scale to 221,741 recipes.

2 / Agentic orchestration improves flexibility only when tool contracts stay precise.

3 / Grounding and independent validation materially reduce hallucination risk.

4 / Debugging state and argument flow is core AI engineering—not cleanup.

A rigorous engineering investigation into real-world AI deployment challenges.



Thank you for your attention